Initial conditions

Purpose

The initial-conditions files set the starting state of land and water objects before the first simulation day. The general files are:

- `plant.ini`. Named plant communities. One community is a list of plants with initial LAI, biomass, heat-unit accumulation, plant population, fraction of years to maturity, and residue cover. Land-use entries in `landuse.lum` point at a community name through the `plant_cov` field.
- `soil_plant.ini`. Named soil and plant initial-state records combining a starting soil-water fraction with names of nutrient, pesticide, pathogen, salt, heavy-metal, and constituent initial-condition records. `hru-data.hru` references one row of this file per HRU through `soil_plant_init`.
- `om_water.ini`. Initial organic matter and water properties for water bodies (aquifers, channels, ponds, reservoirs, wetlands). One row per named initialisation record.

The solute-specific initial-condition files (`pest_hru.ini`, `pest_water.ini`, `path_hru.ini`, `path_water.ini`, `hmet_hru.ini`, `hmet_water.ini`, `salt_hru.ini`, `salt_water.ini`, `cs_hru.ini`, `cs_aqu.ini`, `cs_channel.ini`) are declared in the same `type input_init` block but documented on the module pages linked at the bottom.

Source

Filename strings: src/input_file_module.f90, `type input_init`.

File	Reader	Type populated
`plant.ini`	src/readpcom.f90	`type plant_community_db` (with `type plant_init_db` array) in `src/plant_data_module.f90`
`soil_plant.ini`	src/soil_plant_init.f90	`type soil_plant_initialize` in `src/hru_module.f90`
`om_water.ini`	src/om_water_init.f90	`type hyd_output` in `src/hydrograph_module.f90`

All three readers skip line 1 (title) and line 2 (header), then read records list-directed. Column positions in the reference examples are cosmetic indentation. If the file is missing or the filename in `file.cio` is `null`, the reader allocates an empty array and continues.

plant.ini

Each plant community is a header record followed by `plt_cnt` plant records.

Community record (3 fields)

Read into `pcomdb(icom)`.

#	Field	Type	Units	Default	Description
1	name	char(40)	none	“frsd_frsd”	community name. Referenced from `landuse.lum` (`plant_cov`)
2	plants_com	int	none	1	number of plant records that follow
3	rot_yr_ini	int	none	1	starting year in the rotation

Plant record (8 fields, repeated `plants_com` times)

Read into `pcomdb(icom)%pl(iplt)` (`type plant_init_db`).

#	Field	Type	Units	Default	Description
1	cpnm	char(40)	none	“frsd”	plant name. Must match a row in `plants.plt`. The reader looks the name up in `pldb` and stores the index in `db_num`. If the name is not found a message is written to `9001` (`diagnostics.out`)
2	igro	char(1)	none	“y”	land cover status. `y` = growing, `n` = not growing
3	lai	real	$m^{2/m}2$	0.0	initial leaf area index
4	bioms	real	kg/ha	0.0	initial above-ground biomass
5	phuacc	real	fraction	0.0	fraction of plant heat units accumulated at simulation start
6	pop	real	plants/m^2	0.0	plant population
7	fr_yrmat	real	fraction	1.0	fraction of current year of growth to years to maturity. Used for perennial trees
8	rsdin	real	kg/ha	10000.0	initial surface residue

Example

From `refdata/Ames_sub1/plant.ini`. First two communities, indentation trimmed for width:

```
plant.ini:
pcom_name   plt_cnt  rot_yr_ini  plt_name  lc_status  lai_init  bm_init  phu_init  plnt_pop  yrs_init  rsd
cosy_comm         2           1
                              corn        n   0.00000      0.00000   0.00000   0.00000   0.00000   10000.0
                              soyb        n   0.00000      0.00000   0.00000   0.00000   0.00000   10000.0
corn_comm         1           1
                              corn        n   0.00000      0.00000   0.00000   0.00000   0.00000   10000.0
```

soil_plant.ini

One row per initialisation record. Rows are referenced by name from `hru-data.hru` through the `soil_plant_init` field.

The number of fields read depends on `bsn_cc%nam1`. When `nam1 == 0` (the default) the reader reads 7 fields. When `nam1 /= 0` it reads an eighth field (`csc`) for the constituent-side initial-condition record. `nam1` is declared in `basin_module.f90` as "not used", so in practice the 7-field form is what is written.

Row (7 fields, or 8 with constituents)

Read into `sol_plt_ini(ii)` (`type soil_plant_initialize`).

#	Field	Type	Units	Default	Description
1	name	char(40)	none	“”	record name. Referenced from **hru-data.hru**
2	sw_frac	real	fraction	0.0	initial soil water content as a fraction of field capacity
3	nutc	char(40)	none	“”	name of an initial nutrient record (links to soil nutrient initialisation)
4	pestc	char(40)	none	“”	name of an initial pesticide record. See pest_hru.ini
5	pathc	char(40)	none	“”	name of an initial pathogen record. See path_hru.ini
6	saltc	char(40)	none	“”	name of an initial salt record. See salt_hru.ini
7	hmetc	char(40)	none	“”	name of an initial heavy-metal record. See hmet_hru.ini
8	csc	char(40)	none	“”	name of an initial constituent record. Read only when **bsn_cc%nam1 /= 0**. See cs_hru.ini

Use `null` for any solute-side column that is not active.

Example

From `refdata/Ames_sub1/soil_plant.ini`:

```
soil_plt_ini
       NAME     SW_FRAC    NUT       PEST   PATH   SALT   HMET
soilplant1         0.5    soilnut1   null   null   null   null
```

om_water.ini

One row per named record. Rows are referenced by name from the connectivity files of water bodies (aquifers in `aqu_read_init.f90`, channels, reservoirs, ponds, wetlands).

Row (19 fields)

The reader calls `read (..., om_init_name(ichi), om_init_water(ichi))`. The first field is the name. The remaining 18 fields fill all members of `type hyd_output` in declaration order.

#	Field	Type	Units	Default	Description
1	name	char(16)	none	“”	record name
2	flo	real	m^3 (or fraction for some objects)	0.0	water volume
3	sed	real	metric tons	0.0	sediment
4	orgn	real	kg N	0.0	organic N (PART_N in writer header)
5	sedp	real	kg P	0.0	organic P (PART_P)
6	no3	real	kg N	0.0	NO3-N
7	solp	real	kg P	0.0	mineral soluble P
8	chla	real	kg	0.0	chlorophyll-a
9	nh3	real	kg N	0.0	NH3
10	no2	real	kg N	0.0	NO2
11	cbod	real	kg	0.0	carbonaceous biological oxygen demand
12	dox	real	kg	0.0	dissolved oxygen
13	san	real	tons	0.0	detached sand
14	sil	real	tons	0.0	detached silt
15	cla	real	tons	0.0	detached clay
16	sag	real	tons	0.0	detached small aggregates
17	lag	real	tons	0.0	detached large aggregates
18	grv	real	tons	0.0	gravel
19	temp	real	deg C	0.0	water temperature

The `flo` field is interpreted as a fraction of principal storage for reservoirs and ponds, and as an absolute volume for channels. The interpretation is made by the consumer (`cal_allo_init.f90` and related routines), not the reader.

Example

From `SWATPlusWaterAllocation/documents/wro_test/om_water.ini`:

```
om_wat_ini Generated from ...
       NAME    VOL     SED  PART_N  PART_P    NO3   SOLP  CHL_A    NH3    NO2  CBN_BOD  DIS_OX   SAND   SIL
 aqu_default     1       0       0       0      2      0      0      0      0        0       0      0
 cha_default   0.1    1000      95       0     70      0      0      5      5        0       0      0
pond_default   0.8     100       1     0.1      1   0.03      0    0.1   0.01        0       0      0
 res_default     1     100       1     0.1      1   0.03      0    0.1   0.01        0       0      0
 wet_default   0.5       0       0       0      2      0      0      0      0        0       0      0
```

Related files

- `hru-data.hru`. Per-HRU pointer into `soil_plant.ini` through `soil_plant_init`.
- `landuse.lum`. Land-use entries point at a `plant.ini` community through `plant_cov`.
- `plants.plt`. Plant parameter database. Every `cpnm` in `plant.ini` must resolve here.
- `aquifer.aqu`, `initial.aqu`. Aquifers reference `om_water.ini` records for initial storage.
- `*.res`, `*.wet`, `sd_channel`. Water bodies reference `om_water.ini` for initial state.
- Solute-side initial conditions: `pesticide module`, `pathogen module`, `metals module`, `salt module`, `cs module`.